

Physics 7B — Final Exam Review Session

Worked Solutions with Teaching Notes

Based on the Spring 2001 Final Exam (R. Packard)

UC Berkeley — Spring 2026

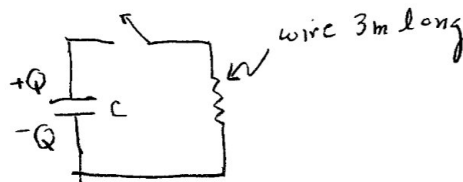
How to use this packet. Each problem is followed by a worked solution and a “Teaching Notes” box that highlights the key physics concept, common student mistakes, and conceptual checkpoints worth emphasizing in section. Numerical answers are boxed in green.

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Problem 1 RC Discharge Through a Resistive Wire (35 pts)

1. (35pts) A capacitor $C=2\text{ }\mu\text{fd}$ is charged with $Q=3\times 10^{-4}\text{C}$ and is connected in series with a wire of length $l=3\text{m}$. The wire has a circular cross-section with radius $r=1\times 10^{-5}\text{m}$, a resistivity $\rho=9.7\times 10^{-8}\Omega\text{m}$, a mass $m=10^{-4}\text{kg}$ and specific heat c_p of 450J/kgC° . When the switch (shown below) is closed at $t=0$ the capacitor discharges through the resistor.
- How long after the switch closes does the voltage on the capacitor reach 75 volts? Call this time t_1 ,
 - At t_1 what is the voltage across the resistance?
 - How much thermal energy is deposited in the wire by the discharge process?
 - What is the temperature rise of the wire due to the complete discharge process? Ignore heat lost to the environment during the discharge process.



Solution

Setup. The wire *is* the resistor. Compute its resistance from geometry and material:

$$R = \frac{\rho \ell}{A} = \frac{\rho \ell}{\pi r^2} = \frac{(9.7 \times 10^{-8} \Omega \text{m})(3 \text{m})}{\pi (1 \times 10^{-5} \text{m})^2} = \boxed{926.28 \Omega}.$$

Part (a) — Time to reach 75 V. The capacitor discharges through R according to the standard RC equation:

$$Q(t) = Q_0 e^{-t/RC}, \quad V_C(t) = \frac{Q_0}{C} e^{-t/RC}.$$

The initial voltage is $V_0 = Q_0/C = (3 \times 10^{-4})/(2 \times 10^{-6}) = 150 \text{V}$. Setting $V_C(t_1) = 75 \text{V}$:

$$\frac{75}{150} = e^{-t_1/RC} \implies t_1 = RC \ln\left(\frac{Q_0}{CV_C(t_1)}\right) = RC \ln 2.$$

Plugging in:

$$t_1 = (926.28)(2 \times 10^{-6}) \ln 2 \approx 1.28 \times 10^{-3} \text{s}.$$

$$t_1 \approx 1.28 \text{ ms}$$

Part (b) — Voltage across R at t_1 . Kirchhoff's voltage law around the single loop demands $V_R + V_C = 0$ during discharge (the EMF source is gone), so

$$|V_R(t_1)| = |V_C(t_1)| = \boxed{75 \text{ V}}.$$

Note: this is *exactly* 75 V, not an approximation from plugging t_1 back in.

Part (c) — Total thermal energy deposited. Energy conservation: every joule initially stored in the capacitor must end up as heat in the wire (the capacitor ends fully discharged):

$$U_{\text{thermal}} = U_{C,\text{initial}} = \frac{Q_0^2}{2C} = \frac{(3 \times 10^{-4})^2}{2(2 \times 10^{-6})} = \boxed{0.0225 \text{ J}}.$$

You can verify by integrating the dissipated power: $\int_0^\infty I^2 R dt$ gives the same number.

Part (d) — Temperature rise. Apply $Q = mc_p \Delta T$:

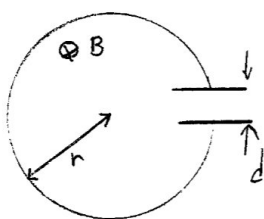
$$\Delta T = \frac{Q}{mc_p} = \frac{0.0225}{(10^{-4})(450)} = \boxed{0.5^\circ \text{C}}.$$

Teaching Notes & Common Pitfalls

- **The wire IS the resistor.** Students often look for a separate resistor symbol. Emphasize that any real conductor with $\rho > 0$ dissipates energy.
- **Initial voltage check.** $V_0 = Q_0/C = 150 \text{ V}$, so reaching 75 V is exactly one “half-life” of the exponential, giving $t_1 = RC \ln 2$. Students who notice this can do part (a) almost in their head.
- **Part (b) trap.** The grading scheme docks a point for plugging t_1 back into the formula. Kirchhoff's law gives the answer *exactly* — always 75 V at t_1 , regardless of how messy the numerical t_1 was.
- **Energy conservation shortcut.** For part (c), don't integrate $I^2 R dt$ unless asked to. The capacitor's stored energy *must* go entirely into the wire.

Problem 2 Faraday's Law and Capacitor Breakdown (20 pts)

2. (20pts) A flat circular loop (radius $r=10\text{cm}$) of wire connects the two sides of a parallel plate capacitor with gap $d=0.1\text{mm}$. The breakdown electric field strength of the air in the gap is $3\times 10^6\text{Nt/C}$. A uniform magnetic field perpendicular to the plane of the loop is increasing exponentially in time: $B = 2e^{3t} (T)$. What is the value of the magnetic field when the gap breaks down?



Solution

Strategy. Time-varying B through the loop induces an EMF, which drives charge until enough builds up on the capacitor plates to create the breakdown E field in the gap.

Step 1 — Induced EMF from Faraday's law. The flux through the loop is $\Phi_B = B \pi r^2$. With $B = 2e^{3t}$,

$$\frac{d\Phi_B}{dt} = \pi r^2 \frac{dB}{dt} = 3B \pi r^2.$$

By Faraday's law, $|\mathcal{E}| = |d\Phi_B/dt| = 3B \pi r^2$.

Step 2 — Where does the EMF appear? This is the key conceptual step. The wire is a conductor, so $\vec{E}_{\text{inside wire}} = 0$. The entire EMF must therefore appear across the only break in the loop — the capacitor gap:

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{\ell} = E_{\text{gap}} d.$$

Step 3 — Apply the breakdown condition. Breakdown happens when $E_{\text{gap}} = E_{\text{breakdown}} = 3 \times 10^6 \text{ V/m}$:

$$E_{\text{breakdown}} d = 3B \pi r^2 \implies B = \frac{E_{\text{breakdown}} d}{3\pi r^2}.$$

Plugging in:

$$B = \frac{(3 \times 10^6)(10^{-4})}{(3) \pi (0.1)^2} \approx \boxed{3.2 \times 10^3 \text{ T}}.$$

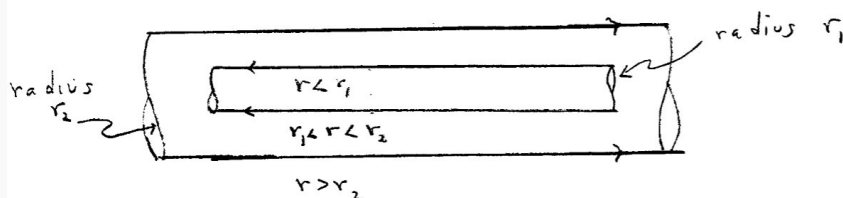
$$\boxed{B_{\text{breakdown}} \approx 3.2 \times 10^3 \text{ T}}$$

Teaching Notes & Common Pitfalls

- **The grading rubric flags one common mistake explicitly:** writing $\oint \vec{E} \cdot d\vec{\ell} = 2\pi r E$. That formula applies to the *induced* field configuration around an axis of symmetry — not to the gap voltage of a capacitor. Here we want the EMF to live entirely in the gap.
- **The induced EMF only “wants” to drive current.** Once enough charge accumulates on the plates to oppose further charging (electrostatically), the gap voltage equals the EMF.
- **Why no t in the answer?** Because we set $E_{\text{gap}} = 3B\pi r^2/d$ at the breakdown moment — B at that moment is what we want. The exponential growth law just tells us *how fast* we get there, not the breakdown value.
- **Sanity check on units:** $3 \times 10^3 \text{ T}$ is enormous (MRI magnets are $\sim 3 \text{ T}$). That’s fine for a textbook problem — the small loop area and large breakdown field demand a huge B .

Problem 3 Coaxial Cable: B Field and Magnetic Energy (20 pts)

3. (20pts) Consider a long cable made of two coaxial thin cylinders. The inner cylinder has radius r_1 and the outer cylinder has radius r_2 . The inner cylinder carries a current $+I$ and $-I$ flows in the outer conductor.
- Compute the magnetic field B for the three regions: $r < r_1$; $r_1 < r < r_2$; and $r > r_2$. Neglect the thickness of the conductors. Do not just write down the answer from your notes. Show the calculation.
 - Use the result in part a. to compute the magnetic energy stored (per unit length) in the cable.



Solution

Part (a) — Magnetic field in the three regions. By cylindrical symmetry, $\vec{B}$ circles the axis and depends only on r . Choose an Ampèrian loop = circle of radius r coaxial with the cable. Then $\oint \vec{B} \cdot d\vec{\ell} = 2\pi r B(r)$, and Ampère's law gives

$$2\pi r B(r) = \mu_0 I_{\text{enc}}.$$

The enclosed current depends on which region we're in:

- Region 1** ($r < r_1$): inside the inner cylinder (a hollow shell), no current is enclosed:

$$B(r < r_1) = 0.$$

- Region 2** ($r_1 < r < r_2$): the loop encloses only the inner cylinder's current $+I$:

$$B(r_1 < r < r_2) = \frac{\mu_0 I}{2\pi r}.$$

- Region 3** ($r > r_2$): loop encloses $+I$ from inside and $-I$ from outside, summing to zero:

$$B(r > r_2) = 0.$$

This is the famous “coaxial cable” result: *the field is confined entirely between the conductors.*

Part (b) — Magnetic energy per unit length. The magnetic energy density is $u = B^2/(2\mu_0)$. Substituting the result from region 2:

$$u(r) = \frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2\pi r} \right)^2 = \frac{\mu_0 I^2}{8\pi^2 r^2}, \quad r_1 < r < r_2.$$

Outside this band $u = 0$.

Use a thin cylindrical shell of radius r , thickness dr , length ℓ as the volume element. Its volume is $dV = 2\pi r \ell dr$, so

$$dU = u dV = \frac{\mu_0 I^2 \ell}{4\pi r} dr.$$

Integrate from r_1 to r_2 :

$$U = \int_{r_1}^{r_2} \frac{\mu_0 I^2 \ell}{4\pi r} dr = \frac{\mu_0 I^2 \ell}{4\pi} \ln\left(\frac{r_2}{r_1}\right).$$

Energy per unit length:

$$\boxed{\frac{U}{\ell} = \frac{\mu_0 I^2}{4\pi} \ln\left(\frac{r_2}{r_1}\right)}.$$

Bonus connection to inductance. Recall $U = \frac{1}{2}LI^2$. Comparing,

$$\frac{L}{\ell} = \frac{\mu_0}{2\pi} \ln\left(\frac{r_2}{r_1}\right),$$

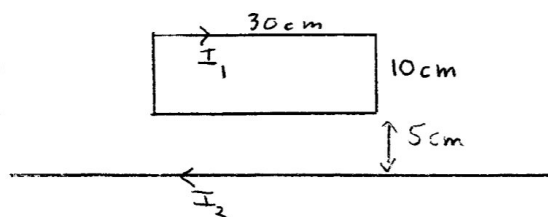
the textbook result for the self-inductance per unit length of a coaxial cable.

Teaching Notes & Common Pitfalls

- **The thin-shell volume element trick.** Many students try Cartesian coordinates and get stuck. Whenever the integrand depends only on r , use cylindrical (or spherical) shells.
- **The vanishing exterior field.** Stress this: a coaxial cable is its own shielding because $B = 0$ for $r > r_2$. This is the whole reason coax exists for signal transmission.
- **Don't write the answer from memory.** The problem explicitly demands showing the Ampère's law calculation. Students who quote $B = \mu_0 I / (2\pi r)$ from the textbook lose points.
- **Connecting to inductance** is a great elegant aside if time permits — it shows the same answer two ways and builds intuition for energy-stored-in-fields vs. energy-stored-in-circuit-elements.

Problem 4 Force & Torque on a Current Loop (20 pts)

4. (20pts) A rectangular loop of wire with the dimensions shown below carries a current $I_1=3$ amps. The loop lies in a plane of a very long straight wire carrying a current $I_2=22$ A.
- Is there a torque on the loop? If your answer is yes compute the torque (magnitude and direction)
 - Is there a net force on the loop? If your answer is yes, compute the force (magnitude and direction).



Solution

Geometry setup. The long wire $I_2 = 22$ A runs horizontally below the loop. The rectangle is 30 cm wide $\times$ 10 cm tall, with its bottom edge 5 cm above the wire. So:

- Bottom edge of loop: distance $r_1 = 5$ cm = 0.05 m from I_2 .
- Top edge of loop: distance $r_2 = 15$ cm = 0.15 m from I_2 .

By the right-hand rule (with I_2 pointing left as drawn), $\vec{B}$ from the long wire points *into the page* everywhere in the loop's plane.

Part (a) — Torque on the loop. The loop's magnetic dipole moment $\vec{\mu} = I_1 \vec{A}$ also points into the page (by the right-hand rule applied to I_1 's circulation). Since $\vec{\mu} \parallel \vec{B}$,

$$\vec{\tau} = \vec{\mu} \times \vec{B} = \boxed{\vec{0}}.$$

No torque.

Part (b) — Net force on the loop. This is the subtle part: $\vec{\tau} = 0$ does *not* mean $\vec{F} = 0$. Why? Because $\vec{B}$ is non-uniform — it varies as $1/r$. The two horizontal sides experience equal forces in opposite directions, but at different r :

- **Two vertical sides (left & right):** forces are equal and opposite by symmetry — they cancel.
- **Top side (length $\ell = 0.30$ m, at $r_2 = 0.15$ m):** carries current opposite to I_2 (judging from the diagram), so by $\vec{F} = I\vec{\ell} \times \vec{B}$ the force points *away from* I_2 — i.e., **upward**.

$$F_{\text{top}} = I_1 \ell \frac{\mu_0 I_2}{2\pi r_2} \quad (\text{up}).$$

- **Bottom side (length ℓ , at $r_1 = 0.05$ m):** carries current in the same direction as I_2 , so the force is attractive — **downward (toward I_2)**.

$$F_{\text{bot}} = I_1 \ell \frac{\mu_0 I_2}{2\pi r_1} \quad (\text{down}).$$

Since $r_1 < r_2$, the bottom force is larger. The net force points downward (toward the wire):

$$F_{\text{net}} = \frac{\mu_0 I_1 I_2 \ell}{2\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right).$$

Plugging in:

$$F_{\text{net}} = \frac{(4\pi \times 10^{-7})(3)(22)(0.30)}{2\pi} \left(\frac{1}{0.05} - \frac{1}{0.15} \right).$$

The bracket: $20 - 6.67 = 13.33$ 1/m. Then:

$$F_{\text{net}} = (2 \times 10^{-7})(3)(22)(0.30)(13.33) \approx 5.28 \times 10^{-5} \text{ N}.$$

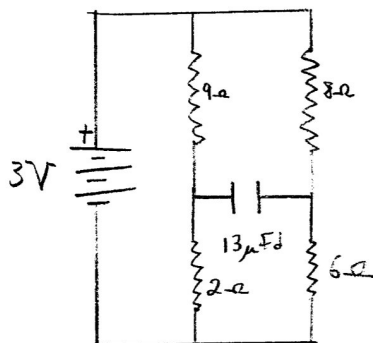
$$\vec{F}_{\text{net}} \approx 5.28 \times 10^{-5} \text{ N, directed downward (toward the long wire).}$$

Teaching Notes & Common Pitfalls

- **The big lesson:** $\vec{\tau} = \vec{\mu} \times \vec{B}$ assumes **uniform** $\vec{B}$. For non-uniform fields, you must evaluate $\vec{F} = I\vec{\ell} \times \vec{B}$ on each segment. The dipole formula handles *rotation tendency* only when the field is uniform across the loop.
- **Force from current carried by a single wire:** parallel currents attract, antiparallel repel. Use this as a quick sign check before doing the cross products.
- **Why does the loop get pulled toward the long wire?** Because the closer (bottom) side feels the stronger attractive force. This is the same mechanism behind why a paramagnet is pulled into a field-gradient region.
- **Common mistake:** forgetting that the side currents (vertical legs) cancel only by symmetry — be sure to point out the $1/r$ field is the same magnitude on left and right at any given height.

Problem 5 Steady-State Energy Stored in a Capacitor (15 pts)

5. (15pts) In the circuit shown below, find the energy stored on the capacitor when it is fully charged.



Solution

Key insight. “Fully charged” means $t \rightarrow \infty$. In steady state, **no current flows through the capacitor** (the capacitor is an open circuit at DC). Therefore the circuit splits into two independent branches:

- **Branch A:** battery $\rightarrow 9\Omega \rightarrow 2\Omega \rightarrow$ back to battery. Total = 11Ω .
- **Branch B:** battery $\rightarrow 8\Omega \rightarrow 6\Omega \rightarrow$ back to battery. Total = 14Ω .

Currents:

$$i_1 = \frac{3\text{ V}}{11\Omega} \approx 0.273\text{ A}, \quad i_2 = \frac{3\text{ V}}{14\Omega} \approx 0.214\text{ A}.$$

Voltages at the capacitor terminals. Let node A be the junction between the 9Ω and 2Ω resistors (one capacitor plate), and node B the junction between the 8Ω and 6Ω resistors (other plate). Taking the negative terminal of the battery as our 0 V reference:

$$V_A = 3\text{ V} - i_1 \cdot 9\Omega = 3 - (0.273)(9) \approx 0.55\text{ V}.$$

$$V_B = 3\text{ V} - i_2 \cdot 8\Omega = 3 - (0.214)(8) \approx 1.29\text{ V}.$$

Voltage across the capacitor.

$$V_C = |V_A - V_B| = |0.55 - 1.29| \approx 0.74\text{ V}.$$

Energy stored.

$$U = \frac{1}{2}CV_C^2 = \frac{1}{2}(13 \times 10^{-6}\text{ F})(0.74)^2 \approx \boxed{3.56 \times 10^{-6}\text{ J}}.$$

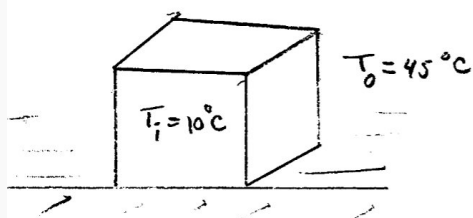
$$\boxed{U \approx 3.56\text{ }\mu\text{J}}$$

Teaching Notes & Common Pitfalls

- **Capacitor at DC steady state = open circuit.** This is the single most important pedagogical point of this problem. Once students internalize this, the problem becomes a routine voltage-divider exercise.
- **Don't average!** A common mistake is to combine the resistors using parallel/series tricks that ignore the fact the two branches are coupled *only* through the (open) capacitor. Stress that the two branches operate independently in steady state.
- **Sign of V_C doesn't matter for energy** because $U = \frac{1}{2}CV^2$ is even in V . But for charge, $Q = CV$, sign matters — the higher-potential side (B here) accumulates positive charge.
- **Order of operations on the partial credit:** the rubric awards 4 pts just for drawing the equivalent “ $t = \infty$ ” circuit. Encourage students to draw it explicitly even if they're sure of the answer.

Problem 6 Heat Pump Cooling Cost (30 pts)

6. (30pts) A cubical food storage container (side length $L=3\text{m}$) has walls and roof of thickness $d=0.1\text{ m}$. The walls and roof are made of material with thermal conductivity $\kappa=0.1\text{J/smC}^\circ$. On a hot day the outside temperature is 45°C . The owner of the food wants to keep the inside of the container at 10°C by running an electrically powered heat pump cooling system. The heat pump runs at 70% of the Carnot efficiency. If the cost of electricity is $\$0.30/\text{kw-hr}$, how much does it cost to cool the container for ten hours? Ignore the heat loss through the floor.



Solution

Roadmap. Three sub-calculations:

1. Heat leak rate $\dot{Q}_L$ through walls + roof (Fourier's law).
2. Heat-pump coefficient of performance (COP) at 70% of Carnot.
3. Electric power needed = (heat to remove) / COP, then convert to dollars.

Step 1 — Heat conduction through walls + roof. The container is a cube with side $L = 3\text{ m}$. Ignoring the floor leaves 4 walls + 1 roof = 5 surfaces, each of area L^2 . Total area:

$$A = 5L^2 = 5(3)^2 = 45\text{ m}^2.$$

Fourier's law (steady-state conduction through slab of thickness d):

$$\frac{Q}{t} = \kappa \frac{A}{d} \Delta T = (0.1) \frac{45}{0.1} (45 - 10) = (0.1)(450)(35) = 1575\text{ W}.$$

Step 2 — COP of the heat pump. For cooling (a refrigerator, not a heater), the figure of merit is

$$\text{COP}_{\text{cool}} = \frac{Q_L}{W} = \frac{Q_L}{Q_H - Q_L}.$$

For an ideal Carnot refrigerator,

$$\text{COP}_{\text{Carnot}} = \frac{T_L}{T_H - T_L}.$$

Convert to Kelvin: $T_L = 283\text{ K}$, $T_H = 318\text{ K}$, so $T_H - T_L = 35\text{ K}$. Then

$$\text{COP}_{\text{Carnot}} = \frac{283}{35} \approx 8.09.$$

At 70% of Carnot:

$$\text{COP} = 0.70 \times 8.09 \approx 5.66.$$

Step 3 — Electrical power. To keep the inside at 10°C in steady state, the pump must extract heat at the rate it leaks in: $\dot{Q}_L = 1575\text{ W}$. So

$$P_{\text{elec}} = \frac{\dot{Q}_L}{\text{COP}} = \frac{1575}{5.66} \approx 278.3\text{ W}.$$

Step 4 — Cost over 10 hours.

$$E = (0.2783\text{ kW})(10\text{ hr}) = 2.783\text{ kWh}.$$

$$\text{Cost} = 2.783\text{ kWh} \times \$0.30/\text{kWh} \approx \boxed{\$0.83}.$$

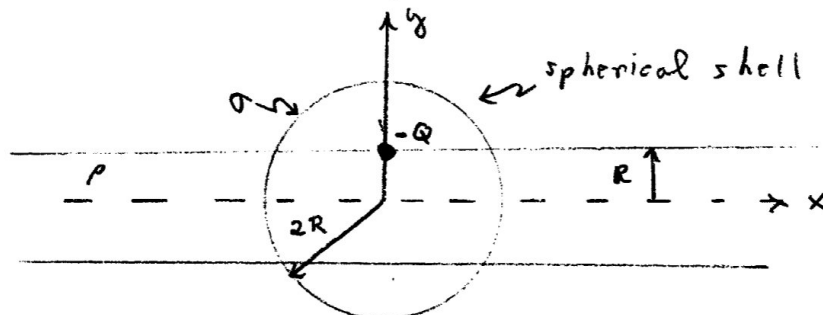
$$\boxed{\text{Cost for 10 hours of cooling} \approx \$0.83}$$

Teaching Notes & Common Pitfalls

- **COP > 1 is not a violation of energy conservation.** A heat pump uses W of electrical work to *move* $Q_L = \text{COP} \cdot W$ joules of thermal energy. The total dumped to the hot reservoir is $Q_H = Q_L + W$. Worth drawing the heat-pump diagram on the board.
- **COP for cooling vs. heating.** For a heater, $\text{COP} = Q_H/W = T_H/(T_H - T_L)$. Always 1 unit larger than the cooling COP. Students mix these up constantly.
- **Always convert to Kelvin** before plugging into Carnot. ΔT is the same in $^\circ\text{C}$ and K, but the absolute temperature in the numerator is not!
- **The five-faces detail.** Ignoring the floor is given in the problem — but emphasize you'd otherwise add $6L^2$. If they forget, their answer is off by a factor of 5/6.
- **Real-world reality check.** 83 cents for 10 hours of cooling a 27 m^3 fridge in 45°C ambient is plausibly cheap because the insulation is decent ($d = 10\text{ cm}$, $\kappa = 0.1$). Drop κ by $10\times$ and the bill grows by $10\times$.

Problem 7 Charge Released near a Cylinder + Spherical Shell (25 pts)

7. (25pts) The x axis forms the symmetry axis of an infinitely long cylinder of radius R carrying a uniform positive charge density ρ . A shell with uniform surface charge density σ and radius $2R$ is centered on the origin. A charge $-Q$ and mass m is released from the point $x=0, y=R$. Find its speed v when it reaches the origin.



Solution

Setup. The point charge $-Q$ starts at rest at $(x, y) = (0, R)$ — which is exactly on the surface of the infinite cylinder. The spherical shell of radius $2R$ is centered at the origin. The charge is released and reaches the origin (which is on the cylinder’s axis).

Step 1 — The spherical shell does no work. By the shell theorem (Gauss's law for spherical symmetry), the E -field *inside* a uniformly charged spherical shell is zero. Both the start point ($y = R < 2R$) and the end point (the origin) lie inside the shell, so the shell exerts no force on the charge along the entire trajectory. **Ignore the shell.**

Step 2 — E -field inside the cylinder. For the infinite cylinder of uniform volume charge density ρ , use a Gaussian cylinder of radius $r < R$ and length L coaxial with the x -axis:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0} \implies E(r) \cdot 2\pi r L = \frac{\rho \pi r^2 L}{\epsilon_0}.$$

Solving:

$$E(r) = \frac{\rho r}{2\epsilon_0} \quad (r < R),$$

pointing radially outward from the cylinder's axis (i.e., in the $+y$ direction at the starting point).

Step 3 — Potential difference from $y = R$ to $y = 0$. The work done by the electric force on $-Q$ as it moves from $y = R$ to $y = 0$ is $W = (-Q) \Delta V$, where $\Delta V = V(0) - V(R)$. Note: $V(0) > V(R)$ since the cylinder is positively charged and the axis is the minimum-distance point inside.

Wait — the axis is the *lowest* potential? Let's check: E points outward from the axis, so V decreases as you move outward. So $V(0) > V(R)$. With $\Delta V = V(R) - V(0) < 0$ being negative if measured “inward,” it's cleaner to compute the magnitude:

$$|V(R) - V(0)| = \int_0^R E(r) dr = \int_0^R \frac{\rho r}{2\epsilon_0} dr = \frac{\rho R^2}{4\epsilon_0}.$$

$$\text{So } V(0) - V(R) = +\frac{\rho R^2}{4\epsilon_0}.$$

Step 4 — Conservation of energy. The negative charge gains kinetic energy moving from *higher* potential energy to *lower*. Its potential energy is $U = qV = (-Q)V$. So:

$$U_i = (-Q)V(R), \quad U_f = (-Q)V(0), \quad U_f - U_i = (-Q)[V(0) - V(R)] = -Q\frac{\rho R^2}{4\epsilon_0} < 0.$$

The PE decreased $\Rightarrow$ KE increased. Energy conservation:

$$\frac{1}{2}mv^2 = -(U_f - U_i) = Q\frac{\rho R^2}{4\epsilon_0}.$$

Solving for v :

$$v = \sqrt{\frac{Q\rho R^2}{2m\epsilon_0}} = R\sqrt{\frac{Q\rho}{2m\epsilon_0}}.$$

$$v = R\sqrt{\frac{Q\rho}{2m\epsilon_0}}$$

Teaching Notes & Common Pitfalls

- **Why the shell drops out:** this is a dressed-up application of the shell theorem. Make sure students explicitly write “ $E_{\text{shell}} = 0$ inside” rather than handwave it. The grading rubric makes this the first step.
- **Sign carefully.** A negative charge in a positively-sourced field is *attracted* to high-charge-density regions, so it should accelerate toward the axis. If your final v^2 comes out negative, you flipped a sign in the potential.
- **The radial vs. axial distinction:** the cylinder's axis is the x -axis, but the motion is in the y -direction. The E field has cylindrical symmetry about the x -axis, so within the cylinder, E depends only on the perpendicular distance from the x -axis — i.e., on $|y|$ for points in the xy -plane. The trip from $y = R$ to $y = 0$ is purely radial in the cylinder's coordinates.
- **Gravity is explicitly ignored** per the official rubric. Students sometimes overcomplicate with mg terms.
- **Sanity check on the trajectory:** once the charge crosses the axis, it overshoots and oscillates. The problem asks only for the speed at the origin, where KE is maximum. (Bonus discussion if you have time: the motion is SHM since $E \propto r$.)

End of review packet — good luck on the final!